ENGINEERING MECHANICS II

<u>MAJOR TOPICS</u>

Chapter One:
Kinematics of a Particle: Rectilinear Kinematics

Chapter Two:
Kinematics of a Particle: Curvilinear Motion

Chapter Three:
Kinetics of a Particle: Force and Acceleration

Chapter Four:
Kinetics of a Particle: Work and Energy

Chapter Five:
Basic Planar Kinematics of a Rigid Body

Chapter Six:
Planar Kinetics of a Rigid Body: Force and Acceleration

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January, 2024

MODULE DESCRIPTION

Year 2, Semester 1

Module Title: **ENGINEERING MECHANICS II**

Code	I3641NM
NQF Level	6
Notional Hours	80
Contact Hours	2 Lectures + 1 Tutorial or 1PS /Week
NQF Credits	8
(Co-requisites)	(I3532NM Engineering Mechanics I)
Prerequisite	None
Compulsory/Elective	Compulsory

Student Assessment Strategies

1. Students will be assessed through continuous assessments activities and a final examination
2. The final mark will be made of 50% Continuous Assessment and 50% Examination (1 x 2-hour paper).
3. The Continuous Assessment will be made up of the following assessment activities:
   (i) Assignments (At least 3 assignments): 40%
   (ii) Tests (At least 2 tests): 60%

Criteria for qualifying for the Examination

To qualify for the exam, a student must obtain a minimum of 50% in the Continuous Assessment.

Criteria for passing the module

To pass this course a student should obtain a minimum exam mark of 40% and average of 50% from both the Continuous Assessment and Examination.

Module Purpose
The purpose of this module is to equip students with fundamental principles and formulations of the kinematics and kinetics of Newtonian mechanics in the context of problems involving the dynamics of particles and rigid bodies.
Main Learning Outcome
Apply the fundamental principles and formulations to solve the kinematics and kinetics of Newtonian mechanics.

Specific Learning Outcomes
On completing the module students should be able to: 1. Competently express motion of a body in terms of position, velocity and acceleration. 2. Apply principles of kinematics and kinetics to describe motion and causes of motion. 3. Solve dynamics problems using rectangular, normal and tangential, polar and cylindrical coordinates. 4. Analyse linear, curvilinear, angular, rotational, projectile and relative motion of particles and systems thereof. 5. Apply equations of motion in rectilinear and plane curvilinear motion. 6. Apply the work-energy principle and impulse-momentum principle to solve particle dynamics problems. 7. Analyse kinetics of a system of particles using the work-energy principle and the impulse-momentum principle.

Module Content

Particle Dynamics: Kinematics of particles: Laws of motion, displacement, velocity, acceleration. Rectilinear Motion, rectangular coordinates. Plane curvilinear motion: normal, tangential and polar coordinates. Constrained motion of connected particles. Motion relative to translating axes, Motion relative to rotating axes. General relative motion. Projectiles. Angular motion. Cylindrical coordinates systems. **Kinetics of particles:** Newton's Second Law of Motion. Equations of motion and their solutions for rectilinear and plane curvilinear motion. Work-energy principle. Power and efficiency. Conservation of energy. Principle of linear impulse and momentum. Angular momentum. **Kinetics of a system of particles**. Generalized Newton's Second Law. Work-energy principle. Impulse-momentum principle.

Contribution to Exit Level Outcome:

1. Problem Solving (Course Outcomes 3, 4, 5 and 6)
2. Application of Scientific and Engineering Knowledge (Course Outcomes 2, 5 and 6)
3. Engineering Methods, Skills, and Tools information Technology (Course Outcomes 3, 4, 6 and 7)

Learning and Teaching Strategies/Activities

The course will be facilitated through the following learning activities:

1. Two lecture periods per week for 14 weeks
2. One tutorial or one practical session per week for 14 weeks
3. Face to face consultations
4. Laboratory activities

Learning and Teaching Enhancement Strategies

1. Internal and external moderation of examination papers and scripts.
2. Peer-review of course outlines and teaching.

3. Student evaluations.
4. Regular review of course content.
5. Effective and efficient supervision and monitoring of assignments, tests and examination.

Mechanisms that will be put in place to monitor student progress, evaluate course impact and effect improvement:

1. One-on-one consultations with students during consultation hours.
2. Allocation of extra reading material where applicable.
3. Implement a bi-semester course evaluation to be administered through a google survey.

References

[1] R.C. Hibbeler, Engineering Mechanics: Dynamics (2016), 14th Edition or later, Prentice Hall Publishers

[2] Andrew Pytel and Jaan Kiusalaas, Engineering Mechanics: Dynamics (2016), Brooks & Cole Publishing Company.